

RETENNET MODEL PERFORMANCE

Executive Summary & Technical Evaluation Metrics

1. Executive Summary

Following comprehensive feature engineering and rigorous hyperparameter tuning, the RetenNet Support Vector Machine (SVM) model has achieved production-ready status. The integration of mathematically robust sliding windows and business-logic injections (e.g., distinguishing "Passive Savers" from "True Abandoners") successfully mitigated historical false-positive alarms.

Strategic Business Win

The tuned model successfully reduced False Positives from over 4,200 to approximately 1,000 users. This refinement saves the enterprise from wasting retention marketing budgets on ~3,200 safe customers, while maintaining an exceptional 94% detection rate for actual churners.

2. Key Performance Indicators (KPIs)

0.957

ROC-AUC SCORE

94%

CHURN RECALL

58%

CHURN PRECISION

0.72

F1-SCORE

3. Tuned Classification Report

The table below details the performance metrics across the validation dataset. The critical metric—Precision for Class 1 (Churn)—surged from a baseline of 0.26 up to 0.58 post-tuning, reflecting a dramatically tightened decision boundary.

CLASS	PRECISION	RECALL	F1-SCORE	SUPPORT
Active Users (0)	0.99	0.90	0.94	10,367
Churners (1)	0.58	0.94	0.72	1,523
Macro Avg	0.79	0.92	0.83	11,890

4. Explainability & Feature Impact

Using SHAP (Shapley Additive Explanations) and Permutation Importance, the model's "black box" was decoded to provide transparent intelligence for the Customer Success team. The top algorithmic drivers for churn risk prediction are:

- **Zero-Balance Abandonment (is_true_abandoner):** The strongest leading indicator of total platform exit.
- **Recency Velocity (days_since_last_txn):** Captures the fading momentum of user engagement before absolute drop-off.
- **Capital Retention (is_passive_saver):** Acts as a strong negative-churn anchor, proving that lack of recent transactions does not equate to churn if the user maintains a high balance.
- **Account Tenure:** Prevents false alarms during the initial 14-day user onboarding window.